Zorian Thornton

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Professional Summary

Computational biologist and machine learning researcher with 6+ years of experience developing predictive models for biological systems. Expertise in building end-to-end deep learning pipelines for large-scale sequence analysis, implementing Bayesian statistical methods, and translating complex biological data into actionable insights. Proven track record of developing production-grade tools for viral evolution modeling and protein phenotype prediction.

Technical Skills

Languages: Python, R, SQL, Java, MATLAB, bash
Machine Learning: PyTorch, TensorFlow, JAX, NumPyro, scikit-learn, supervised/unsupervised learning, deep learning, Bayesian inference
Statistical Methods: MCMC, variational inference, hierarchical modeling, dimensionality reduction, model benchmarking
Bioinformatics & Computational Biology: Phylogenetic inference, sequence analysis, deep mutational scanning, Augur, Auspice, Biopython, evofr
Data Science & Visualization: pandas, numpy, scipy, matplotlib, seaborn, statistical modeling
Infrastructure & Tools: Git, SLURM, CUDA, parallel computing, Unix/Linux, high-performance computing

Research & Work Experience

Fred Hutchinson Cancer Research Center — Seattle, WA
Doctoral Researcher, Computational Biology (Advisor: Frederick Matsen IV) — *July 2021 – June 2026*

- Built end-to-end PyTorch pipelines for viral fitness prediction from deep mutational scanning data, processing datasets of tens of thousands of protein variants with genotype-phenotype mapping models
- Developed biophysically-inspired neural network architectures that predict variant effects from experimental measurements, enabling accurate phenotype predictions across multiple viral protein phenotypes
- Created open-source Java simulation framework for modeling viral evolution under immune selection, integrating phylogenetic inference with phenotypic fitness landscapes
- Collaborated with experimental biology teams to validate computational predictions, resulting in 2 peer-reviewed publications and tools adopted by multiple research groups

Adaptive Biotechnologies — Seattle, WA
Machine Learning Computational Biology Intern — *June 2022 – Sept. 2022*

- Implemented and benchmarked semi-supervised learning methods for T-cell receptor specificity classification using high-throughput sequencing datasets containing millions of sequences
- Designed scalable data preprocessing pipelines in Python to handle complex biological sequence data, optimizing memory usage and runtime for production deployment
- Conducted systematic downsampling experiments to characterize model performance across varying data conditions, identifying optimal training set sizes for different classification tasks
- Developed visualization dashboards to communicate technical findings to cross-functional teams including immunologists, clinicians, and product managers

Virginia Tech Department of Statistics — Blacksburg, VA
Undergraduate Researcher (Advisor: Leah Johnson) — *Aug. 2017 – May 2019*

- Developed predictive environmental models integrating temperature-dependent biological parameters with disease transmission dynamics to forecast Bluetongue virus spread across climate scenarios
- Built Bayesian hierarchical models using MCMC methods to quantify uncertainty in environmental-disease relationships, incorporating vector life history traits and host-pathogen interactions

Nielsen — Chicago, IL
Data Science Intern — *June 2018 – Aug. 2018*

- Implemented statistical framework to identify and correct errors in scanned receipt data, building automated pipeline for data quality assurance
- Developed anomaly detection algorithms to flag inconsistencies in consumer purchase data at scale

University of Washington Department of Genome Sciences — Seattle, WA
Undergraduate Researcher (Advisor: Jim Bruce) — *June 2017 – Aug. 2017*

- Built probabilistic models to quantify protein-protein interaction competition in cross-linking mass spectrometry data
- Developed 3D visualization tools for analyzing inter-surface regions of cross-linked proteins, contributing to published proteomics software (Journal of Proteome Research)

Education

University of Washington Seattle, WA
Ph.D. in Genome Sciences *Sep. 2019 – March 2026*

Virginia Tech Blacksburg, VA
B.Sc. Statistics & B.Sc. Computational Modeling and Data Analytics *Aug. 2015 – May 2019*

- Minors in Mathematics and Computer Science

Selected Publications

Thornton, Z., Tran, T., Figgins, M., et al. (2026). Simulating the genetic and antigenic evolution of viruses under selection pressure from host immunity. *Manuscript submitted to Virus Evolution.*

Yu, T.*, **Thornton, Z.***, Hannon, W., et al. (2022). A biophysical model of viral escape from polyclonal antibodies. *Virus Evolution*, 8(2), veac110.

El Moustaid, F., **Thornton, Z.**, et al. (2021). Predicting temperature-dependent transmission suitability of bluetongue virus in livestock. *Parasites & Vectors*, 14(1), pp.1-14.

Keller, A., Chavez, J.D., Eng, J.K., **Thornton, Z.** and Bruce, J.E. (2018). Tools for 3D Interactome Visualization. *Journal of Proteome Research*, 18(2), pp.753-758.

Selected Presentations

Thornton, Z. Predicting Disease Progression of Colorectal Cancer via Self-Organizing Maps. RECOMB 2019, George Washington University, Washington DC, May 2019.

Thornton, Z. Modeling Bluetongue Virus via Markov Chain Monte Carlo Methods. Student Experiential Learning Conference, Virginia Tech, Blacksburg, VA, Apr. 2018.

Thornton, Z. Viewing Molecular Interaction Interfaces Through Directed Computational Methods. Department of Genome Sciences Research Symposium, University of Washington, Seattle, WA, Aug. 2017.

Honors & Awards

National Science Foundation Graduate Research Fellowship Program, Honorable Mention 2020
Mu Sigma Rho, National Honor Society for Statistics 2019
College of Science Dean's Roundtable Scholarship, Virginia Tech 2018–2019
Luther and Alice Hamlett Undergraduate Research Grant, Virginia Tech 2018–2019
Fralin Undergraduate Research Fellowship, Virginia Tech 2017–2018
Eckert Statistics Scholar, Virginia Tech 2016–2019